Differences in Neural Activity when Processing Emotional Arousal and Valence in Autism Spectrum Disorders

INSTITUTE for TRANSLATIONAL RESEARCH in Children's Mental Health
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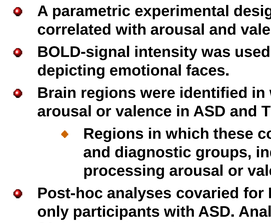
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COLUMBIA PSYCHIATRY

Introduction

- Autism Spectrum Disorders (ASD) are a set of complex neurodevelopmental disabilities that cause lifelong impairments in social ability, communication, and behavioral flexibility (American Psychiatric Association, 2000).
- Individuals with ASD often have difficulty recognizing and interpreting facial emotions, which may impair their ability to understand the intentionality and minds of others, a capacity needed for successful social communication (Grelotti *et al.*, 2002; Golan *et al.*, 2006).
- Numerous studies have provided evidence for the existence of distinct neural systems that subserve the experience of emotional valence (the extent to which emotions are pleasant or unpleasant) and arousal (the extent to which emotions are associated with high or low energy) (Gerber *et al.*, 2008; Posner *et al.*, 2009; Colibazzi *et al.*, 2010).



Graphical representation of the Circumplex Model of Affect with the horizontal axis representing the valence dimension and the vertical axis representing the arousal dimension.

- The Circumplex Model of Affect posits that the subjective experience of all emotion arises from cognitive interpretations of neurophysiological changes in valence and arousal in relation to the experiential context of the eliciting stimuli and memories of prior experiences when similar sensations of valence and arousal are elicited (Russell, 2003; Posner *et al.*, 2005).
- We sought to identify similarities and differences in neural activity when participants with ASD and typically-developing (TD) participants viewed and rated these experiences of facial emotions (Russell et al., 1989; Gerber et al., 2008).
- To our knowledge, this is the first study to assess whether neural activity in circuits that subserve processing of the two dimensions of facial emotions differ between individuals with ASD and their TD counterparts.

Methods

Participant Characteristics

	ASD	TD
Participants (N)	51	84
ASD Subtype:		
PDD-NOS	9	--
Asperger's Syndrome	24	--
Autistic Disorder	18	--
Mean Age (yrs)	27.5 ± 13.1	24.0 ± 11.4
Children (<18yrs) (N / %)	12 (24%)	31 (37%)
Males (N / %)	45 (88%)	62 (74%)
Caucasian (N, %)	40 (78%)	60 (71%)
Mean SES[a]	50	53
Mean FSIQ[b]	109.2 ± 19.4	115.9 ± 12.4
Mean VIQ	110.9 ± 20.9	115.7 ± 13.2
Mean PIQ	105.0 ± 17.6	112.9 ± 11.8
Mean ADOS (Social Affect + Restrictive, Repetitive Behaviors)[c]	10.9 ± 3.1	--
Mean ADOS - Calibrated Severity Scores (Modules 2 and 3)[d]	7.5 ± 1.8	--

[a] SES scores for 7 TD and 14 ASD participants were unavailable.
[b] FSIQ scores for 1 TD participant and 1 ASD participant were unavailable.
[c]ADOS scores for 4 ASD participants were unavailable.
[d]ADOS CSS scores were calculated for participants tested using Modules 2 and 3 (N=10)

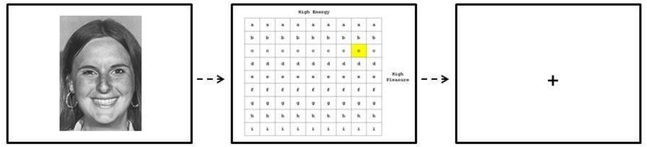
Behavioral Data

- ASD group performed nearly as well as, and in a similar pattern to, the TD group when participants were asked to rate emotional faces for arousal and valence (Tseng et al., 2014).
  - Typical-level behavioral performance on emotion-processing tasks does not exclude the possibility of atypical neurocognitive processing of emotional information.

fMRI Data

- Imaging was performed on a GE Signa 3T whole body scanner (Milwaukee, WI) using a GE single channel quadrature head-coil.
- fMRI data was preprocessed (includding slice timing correction, motion correction, spatial smoothing) using SPM8 (http://www.fil.ion.ucl.ac.uk/spm/, run under MATLAB2009b).
  - Participants for whom motion affected more than 15% of their data were excluded from further analyses (eliminated 1 TD and 4 ASD).
- A parametric experimental design was used to identify brain regions in which neural activity correlated with arousal and valence ratings for a broad range of facial emotions.
- BOLD-signal intensity was used to index neural activity as participants viewed photographs depicting emotional faces.
- Brain regions were identified in which BOLD-signal systematically covaried with ratings of arousal or valence in ASD and TD groups.
  - Regions in which these correlations differed statistically across emotional dimensions and diagnostic groups, indicating the differential associations of these regions with processing arousal or valence within each group.
- Post-hoc analyses covaried for FSIQ in all participants and ADOS scores in analyses involving only participants with ASD. Analyses also conducted using only the adult participants (18-61yrs) in order to detect whether including child participants (7-18yrs) changed our findings. No significant differences from our original findings were detected in these additional analyses.

Arousal and Valence Rating Task



- Each trial consisted of 3 components presented in succession:
  1. Visual presentation of an emotional face for 18 seconds.
  2. Visual presentation of a 2-D grid on which participants indicated their ratings of arousal and valence for each face by moving an arrow controlled by a computer mouse.
  3. Visual presentation of a fixation point (+) at center of participant's visual field.
- Each run consisted of 20 trials presented in a pseudorandom order (but uniform from participant to participant).
- Three runs (totaling 60 stimulus trials) acquired for each person.

Arousal Results

- TD: BOLD-signal correlated inversely with arousal ratings in ILPFC and DLPFC, dorsal ACC, dorsal PC, inferoposterior PC, CN, and PUT.
- ASD: BOLD-signal correlated positively with arousal ratings in posterior temporal and inferior PC, the mesial wall (pregenual and dorsal portions of superior frontal and anterior CG), PreMC and supplementary motor cortices, Cu, PCu, basal ganglia nuclei, THAL, dorsal Cb.
- Correlations of BOLD-signal with arousal ratings differed significantly across ASD and TD participants (ASD>TD) in AMYG, HIPP, CN, ACC, and SFG.

Regions of significant linear correlations of BOLD signal with ratings of AROUSAL for TD participants

Anatomical Regions*	Location Side	BA	MNI Coordinates x	y	z	Z	Correlation
Linear AROUSAL							
Inferolateral Prefrontal Cortex	R		27	62	19	7.29	Negative
Broca's Area	R		51	41	16	6.96	Negative
Posterior Parietal Cortex	R		30	-76	52	6.8	Negative
Dorsolateral Prefrontal Cortex	R		39	56	4	6.74	Negative
Primary Motor Cortex	R		57	11	34	6.74	Negative
Intraparietal Sulcus	L		-44	-46	48	6.73	Negative
Parietal-Occipital-Temporal Cortex	L		-42	-55	61	6.73	Negative
Precuneus	R		12	-70	64	6.34	Negative
Cuneus	R		21	-85	46	6.31	Negative

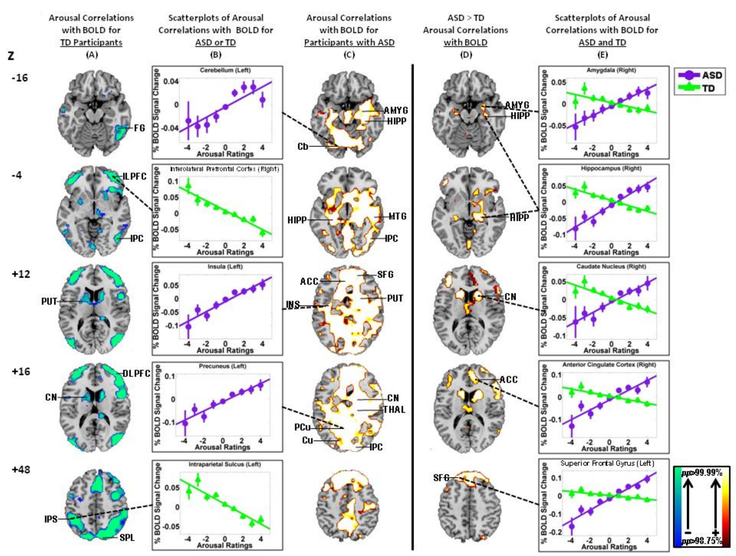
Regions of significant linear correlations of BOLD signal with ratings of AROUSAL for participants with ASD

Anatomical Regions**	Location Side	BA	MNI Coordinates x	y	z	Z	Correlation
Linear AROUSAL							
Premotor Cortex	R	6, 1	39	-28	67	8.21	Positive
Dorsolateral Prefrontal Cortex	L	8	-6	47	25	8.13	Positive
Cerebellum	L		-6	-61	-20	8.04	Positive
Dorsolateral Prefrontal Cortex	L	9	-21	20	58	7.97	Positive
Dorsolateral Prefrontal Cortex	R		21	41	49	7.66	Positive
Broca's Area	L		-51	20	13	4.23	Positive

** Regions were significant at p<0.0125 or Posterior Probability > 98.75%*
*** Regions were significant at p<0.001 or Posterior Probability > 99.0%*

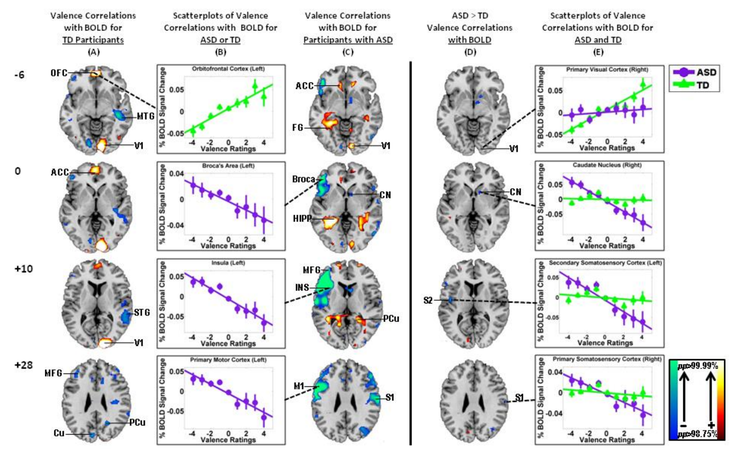
Centers of activation in regions where the linear correlation of BOLD signal intensity with ratings of AROUSAL for participants with ASD differs significantly from the linear correlation of BOLD with ratings of AROUSAL for TD participants.

Anatomical Regions	Location Side	BA	MNI Coordinates x	y	z	Z	Correlation
Linear AROUSAL							
Broca's Area	L	45	-48	41	-5	5.68	ASD>TD
Dorsolateral Prefrontal Cortex	L	9	-24	38	46	5.12	ASD>TD
Dorsolateral Prefrontal Cortex	R	8	12	50	49	5.09	ASD>TD
Broca's Area	R	45	57	26	16	4.95	ASD>TD
Caudate Nucleus	R		18	14	13	4.68	ASD>TD
Hippocampus	R		21	-28	-5	4.56	ASD>TD
Amygdala	R		27	-1	-17	4.41	ASD>TD
Anterior Cingulate Cortex	R	32	12	44	16	4.19	ASD>TD



Valence Results

- BOLD-signal correlated with ratings of linear valence similarly for both diagnostic groups.
- In both TD and ASD participants, valence ratings correlated positively with BOLD-signal in ACC, FG, and, and inversely with BOLD-signal in posterior PC, S1, M1 and SMA.
- Correlations of BOLD-signal with valence ratings did not differ significantly between diagnostic groups.



Discussion

- BOLD-signal correlated linearly with ratings of emotional arousal, but in opposite directions and in differing locations for the two groups.
- BOLD-signal in TD participants correlated inversely with ratings of arousal in regions associated primarily with attentional functions, whereas BOLD-signal in ASD participants correlated positively with arousal ratings in regions most commonly associated with impulse control and with default-mode activity.
- In contrast, BOLD-signal correlated with ratings of valence similarly across the TD and ASD groups, positively in regions associated previously with processing of emotional faces, and inversely in sensorimotor regions.
- Nowhere did the correlation of BOLD-signal with valence ratings differ significantly across groups.
- Our findings provide unique insight into the emotional experiences of individuals with ASD. TD individuals and persons with ASD seem to find differing aspects of emotional stimuli to be salient and relevant. Studying these differences may help us to understand the origins of atypical interpersonal emotional experiences in individuals with ASD.

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